Object Detection I

CS 4391 Introduction Computer Vision

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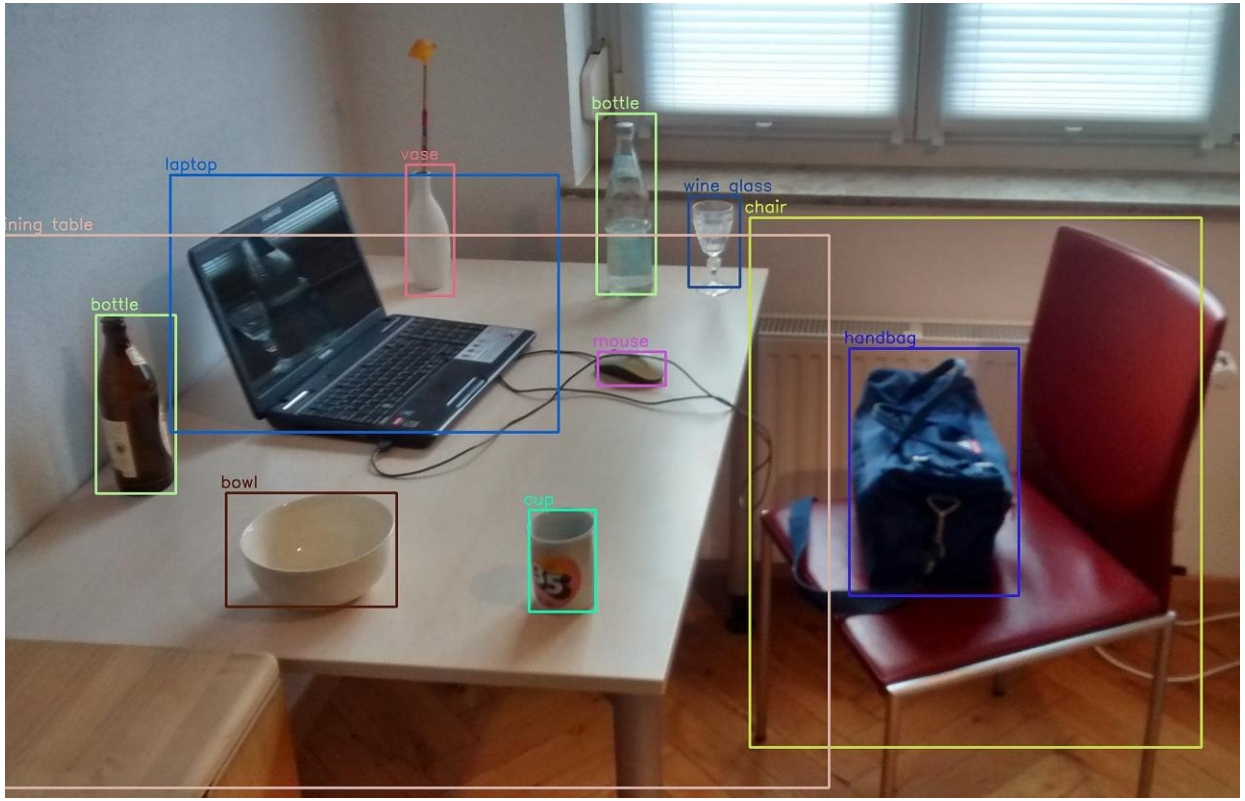
Object Detection

- Localize objects in images and classify them



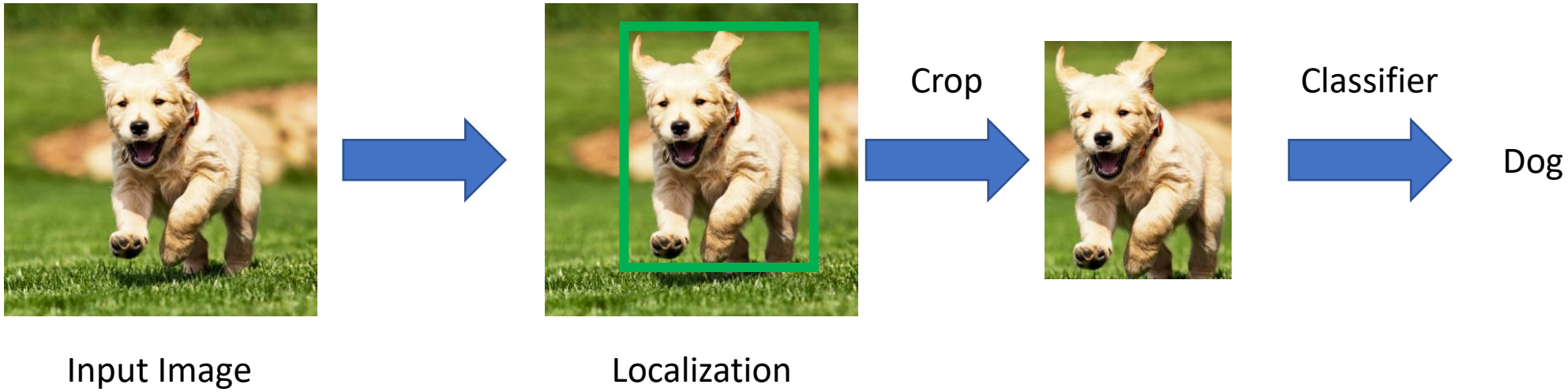
Wikipedia

Why using bounding boxes?

- Easy to store
  - (x, y, w, h): box center with width, height
  - (x1, y1, x2, y2): top left corner and bottom right corner
- Easy for image processing
  - Crop a region

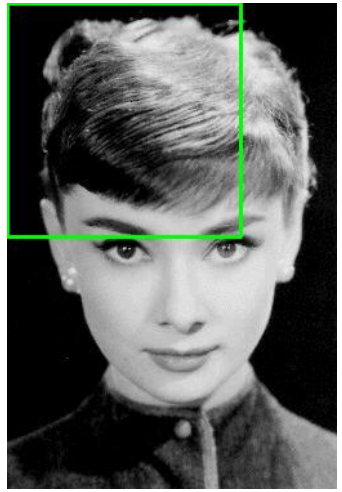
Object Detection

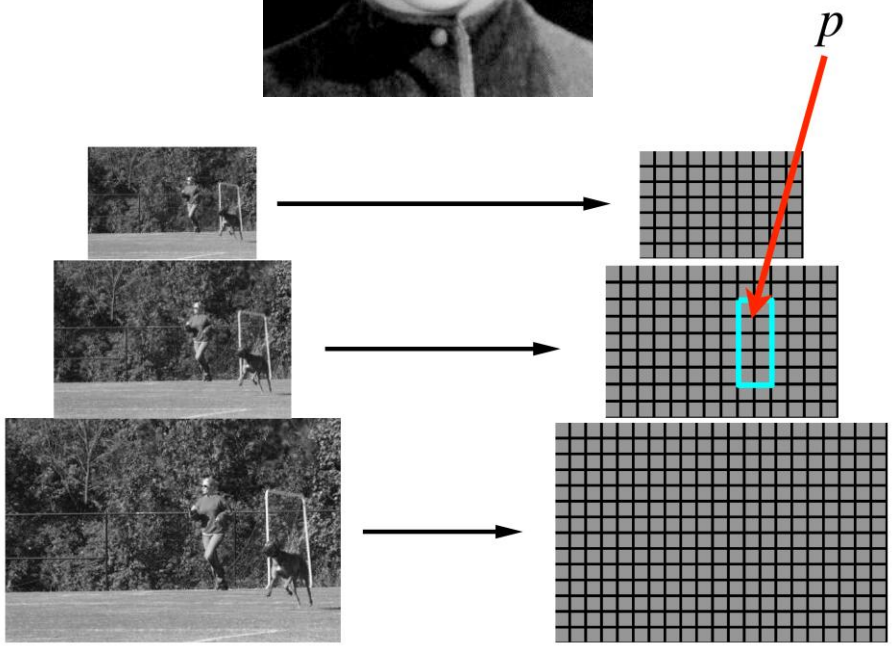
- Localization + Classification



Localization: Sliding Window

- Select a window with a fixed size
- Scan the input image with the window (bounding box)
- How to deal with different object scales and aspect ratios?
  - Use boxes with different aspect ratios
  - Image pyramid

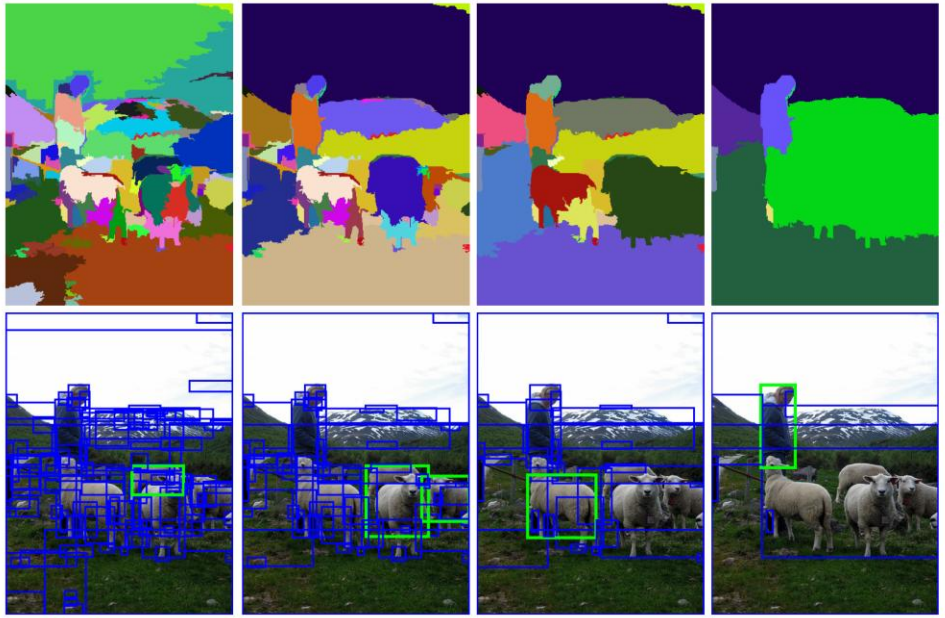




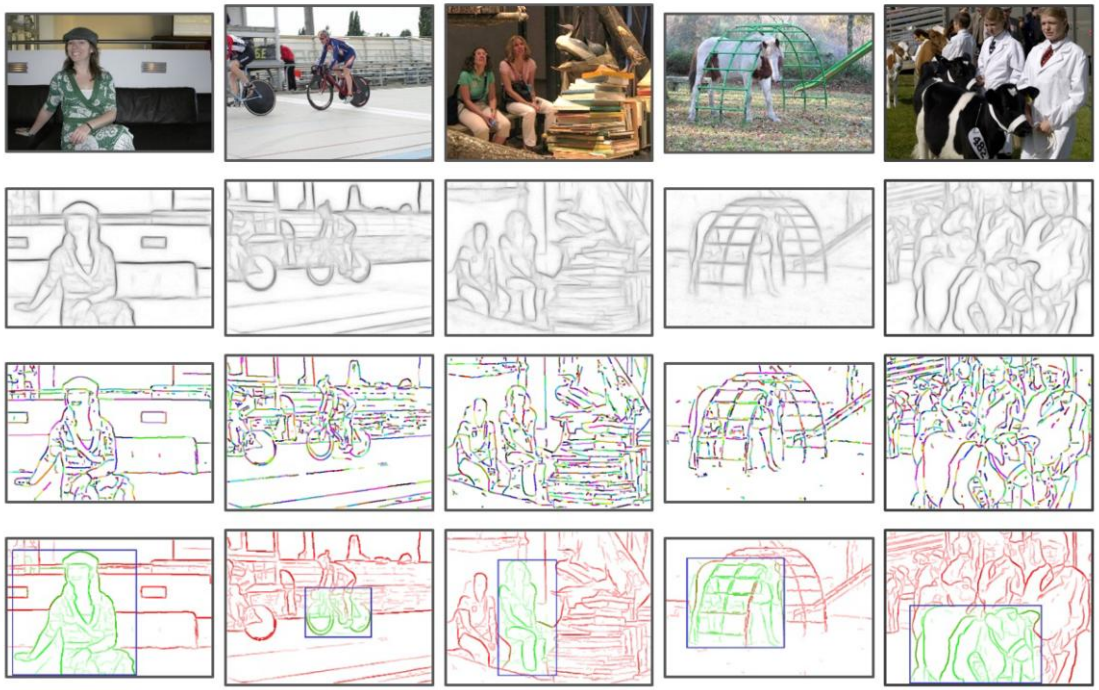
https://cvexplained.wordpress.com/tag/sliding-windows/

Localization: Region Proposal

- Leverage methods that can generate regions with high likelihood of containing objects
  - E.g., bottom-up segmentation methods, using edges



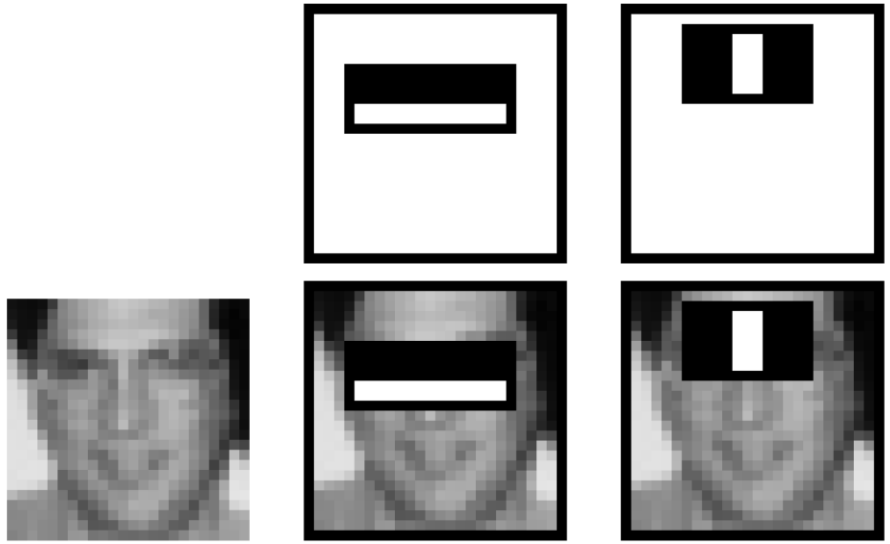
Selective Search, Sande et al., ICCV'11



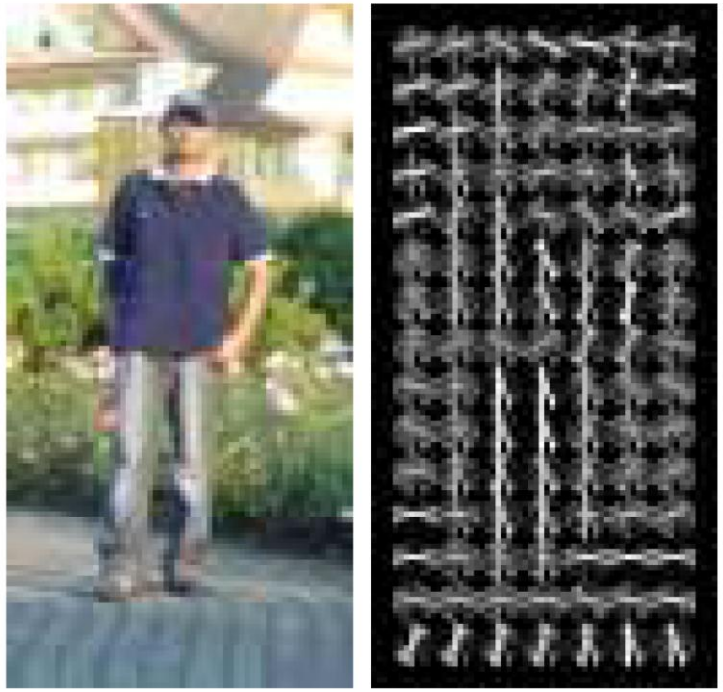
Edge Boxes. Zitnick & Dollar, ECCV'14

Classification: Features

- Traditional methods: Hand-crafted features
- Deep learning methods: learned features in the network



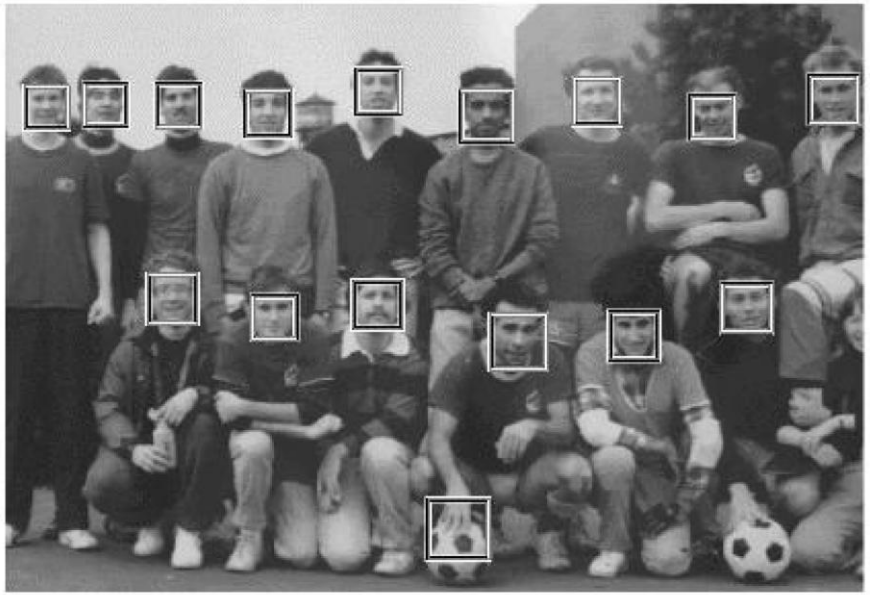
Viola and Jones: rectangle features



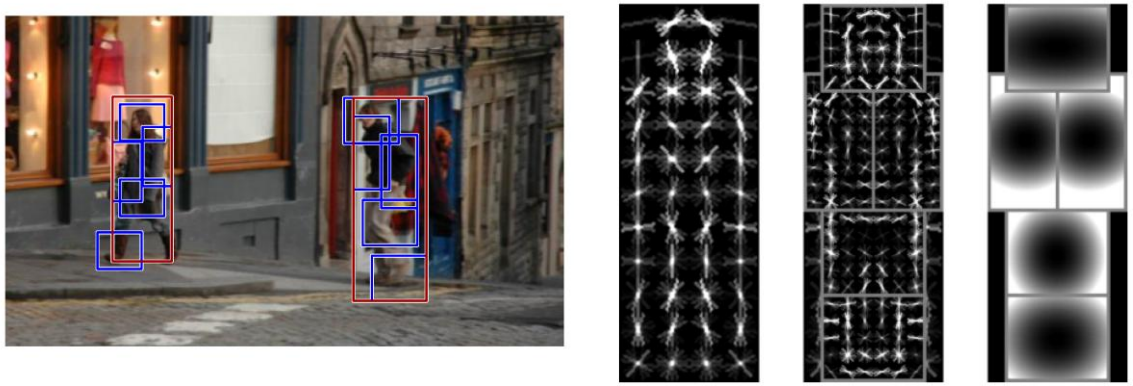
Dadal & Triggs: Histograms of Oriented Gradients

Classification: Classifiers

- Traditional methods
  - AdaBoost
  - Support vector machines (SVMs)
- Deep learning methods
  - Neural networks

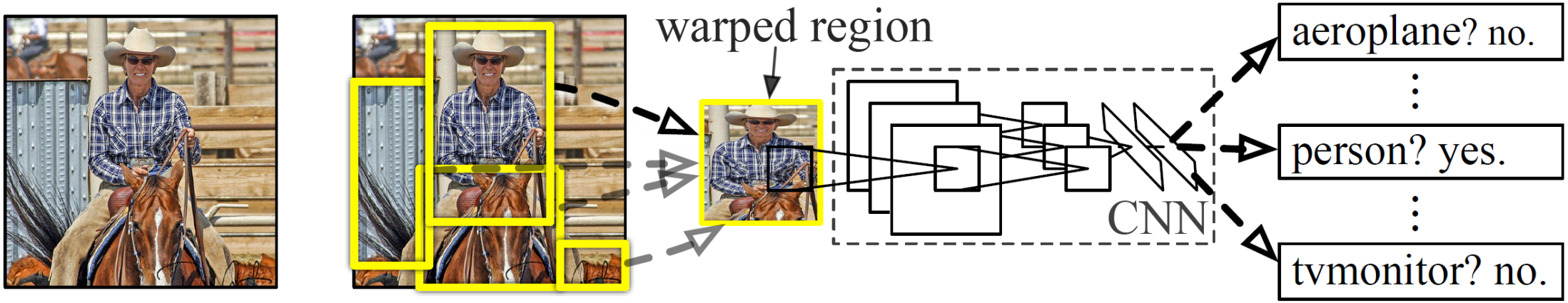


Viola and Jones: AdaBoost
Robust Real-time Object Detection. IJCV, 2001.



Felzenszwalb et al: SVM
Object detection with discriminatively trained part-based models . TPAMI, 2009.

R-CNN



1. Input image

2. Extract region proposals (~2k)

Selective Search

3. Compute CNN features

4. Classify regions

SVM

Rich feature hierarchies for accurate object detection and semantic segmentation. Girshick et al., CVPR, 2014

R-CNN

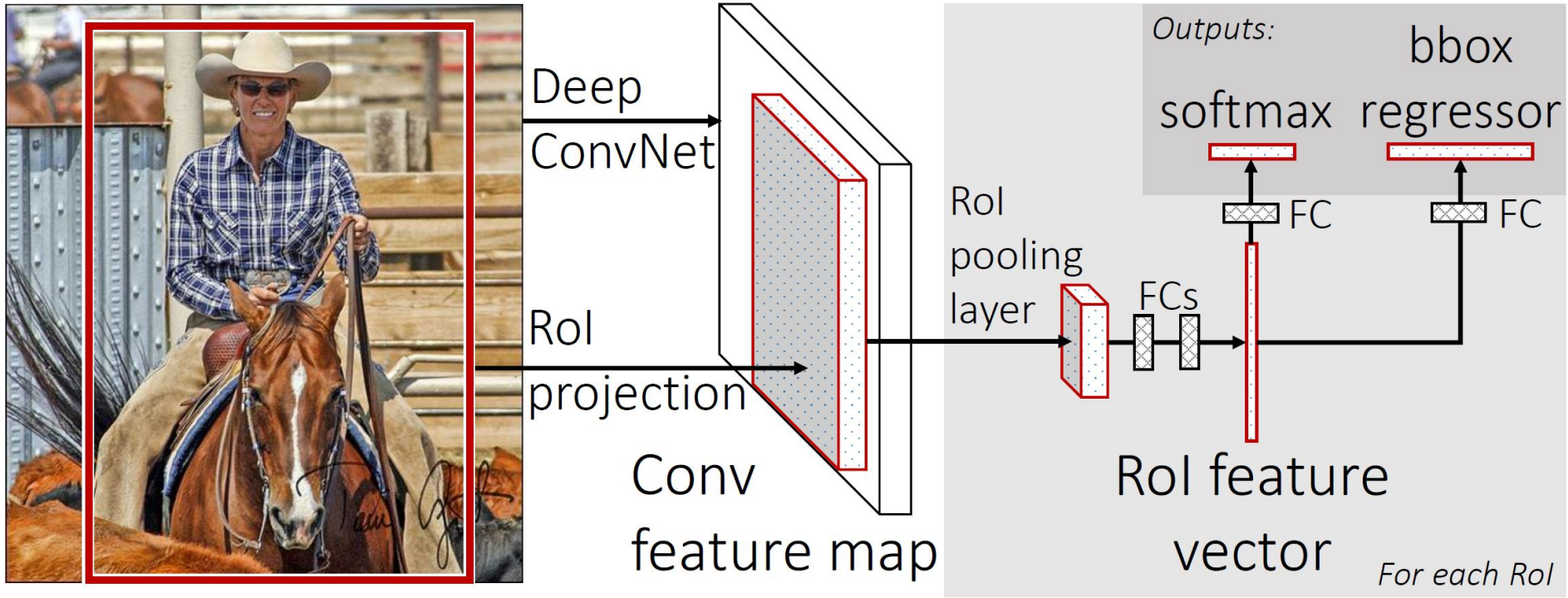
VOC 2007 test	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv	mAP
R-CNN $pool_5$	51.8	60.2	36.4	27.8	23.2	52.8	60.6	49.2	18.3	47.8	44.3	40.8	56.6	58.7	42.4	23.4	46.1	36.7	51.3	55.7	44.2
R-CNN fc_6	59.3	61.8	43.1	34.0	25.1	53.1	60.6	52.8	21.7	47.8	42.7	47.8	52.5	58.5	44.6	25.6	48.3	34.0	53.1	58.0	46.2
R-CNN fc_7	57.6	57.9	38.5	31.8	23.7	51.2	58.9	51.4	20.0	50.5	40.9	46.0	51.6	55.9	43.3	23.3	48.1	35.3	51.0	57.4	44.7
R-CNN FT $pool_5$	58.2	63.3	37.9	27.6	26.1	54.1	66.9	51.4	26.7	55.5	43.4	43.1	57.7	59.0	45.8	28.1	50.8	40.6	53.1	56.4	47.3
R-CNN FT fc_6	63.5	66.0	47.9	37.7	29.9	62.5	70.2	60.2	32.0	57.9	47.0	53.5	60.1	64.2	52.2	31.3	55.0	50.0	57.7	63.0	53.1
R-CNN FT fc_7	64.2	69.7	50.0	41.9	32.0	62.6	71.0	60.7	32.7	58.5	46.5	56.1	60.6	66.8	54.2	31.5	52.8	48.9	57.9	64.7	54.2
R-CNN FT fc_7 BB	**68.1**	**72.8**	**56.8**	**43.0**	**36.8**	**66.3**	**74.2**	**67.6**	**34.4**	**63.5**	**54.5**	**61.2**	**69.1**	**68.6**	**58.7**	**33.4**	**62.9**	**51.1**	**62.5**	**64.8**	**58.5**
DPM v5 [20]	33.2	60.3	10.2	16.1	27.3	54.3	58.2	23.0	20.0	24.1	26.7	12.7	58.1	48.2	43.2	12.0	21.1	36.1	46.0	43.5	33.7
DPM ST [28]	23.8	58.2	10.5	8.5	27.1	50.4	52.0	7.3	19.2	22.8	18.1	8.0	55.9	44.8	32.4	13.3	15.9	22.8	46.2	44.9	29.1
DPM HSC [31]	32.2	58.3	11.5	16.3	30.6	49.9	54.8	23.5	21.5	27.7	34.0	13.7	58.1	51.6	39.9	12.4	23.5	34.4	47.4	45.2	34.3

BB: bounding box regression

Features from AlexNet

Rich feature hierarchies for accurate object detection and semantic segmentation. Girshick et al., CVPR, 2014

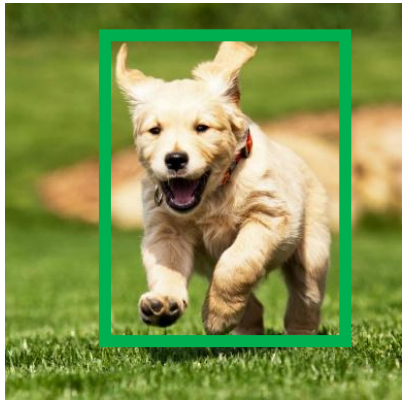
Fast R-CNN



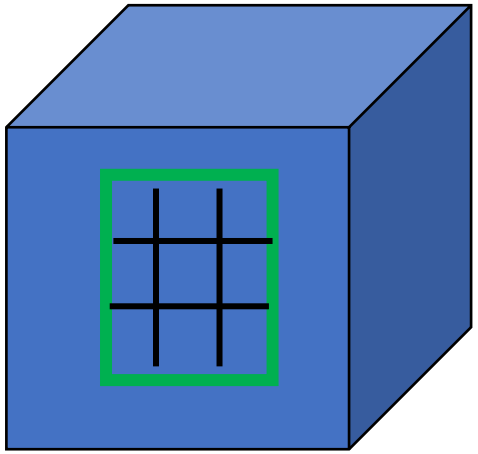
Fast R-CNN. Girshick, ICCV, 2015

RoI Pooling

Divide the mapping RoI into H x W grids

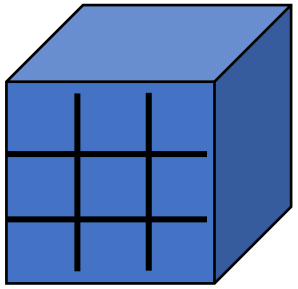


CNN



$$H \times W \times C$$

Max pooling for each grid



RoI

$$(x, y, h, w)$$

RoI mapping to feature map

$$s \times (x, y, h, w)$$

$$s = \frac{1}{16}$$

$$7 \times 7$$

RoI pooling in Fast R-CNN

Bounding Box Regression

- Predict bounding box regression offset for K object classes

$$t^k = (t^k_{\mathrm{x}}, t^k_{\mathrm{y}}, t^k_{\mathrm{w}}, t^k_{\mathrm{h}}) \quad G = (G_x, G_y, G_w, G_h) \quad P = (P_x, P_y, P_w, P_h)$$

Offset

G: ground truth

P: input RoI

$$t_x = (G_x - P_x)/P_w$$
$$t_y = (G_y - P_y)/P_h$$
$$t_w = \log(G_w/P_w)$$
$$t_h = \log(G_h/P_h).$$

$$\hat{G}_x = P_w d_x(P) + P_x$$
$$\hat{G}_y = P_h d_y(P) + P_y$$
$$\hat{G}_w = P_w \exp(d_w(P))$$
$$\hat{G}_h = P_h \exp(d_h(P)).$$

G: ground truth, P: input RoI

Fast R-CNN

- Loss function

Bounding box regress target

$$L(p, u, t^u, v) = L_{\text{cls}}(p, u) + \lambda[u \geq 1] L_{\text{loc}}(t^u, v)$$

Softmax classification probabilities

$$p = (p_0, \ldots, p_K)$$

True class label

Bounding box regress prediction

$$t^u = (t^u_{\text{x}}, t^u_{\text{y}}, t^u_{\text{w}}, t^u_{\text{h}})$$

$$L_{\text{loc}}(t^u, v) = \sum_{i \in \{\text{x,y,w,h}\}} \text{smooth}_{L_1}(t^u_i - v_i)$$

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases}$$

Fast R-CNN

	Fast R-CNN			R-CNN			SPPnet
	S	**M**	**L**	**S**	**M**	**L**	†**L**
train time (h)	**1.2**	2.0	9.5	22	28	84	25
train speedup	**18.3**×	14.0×	8.8×	1×	1×	1×	3.4×
test rate (s/im)	0.10	0.15	0.32	9.8	12.1	47.0	2.3
▷ with SVD	**0.06**	0.08	0.22	-	-	-	-
test speedup	98×	80×	146×	1×	1×	1×	20×
▷ with SVD	169×	150×	**213**×	-	-	-	-
VOC07 mAP	57.1	59.2	**66.9**	58.5	60.2	66.0	63.1
▷ with SVD	56.5	58.7	66.6	-	-	-	-

S: AlexNet, M: VGG, L: deep VGG

SVD for FCs layers

$$W \approx U \Sigma_t V^T$$

Fast R-CNN. Girshick, ICCV, 2015

Summary

- Two-stage detectors
  - R-CNN, Fast R-CNN, Faster R-CNN
  - Region proposal + classification
  - Good performance, slow

- One-stage detectors
  - YOLO, SSD
  - End-to-end network to regress to bounding boxes
  - Fast, comparable performance to two-stage detectors

Further Reading

- Viola–Jones object detection, 2001 https://www.cs.cmu.edu/~efros/courses/LBMV07/Papers/viola-cvpr-01.pdf
- Deformable part model, 2010, https://ieeexplore.ieee.org/document/5255236
- R-CNN, 2014 https://arxiv.org/abs/1311.2524
- Fast R-CNN, 2015 https://arxiv.org/abs/1504.08083
- Faster R-CNN, 2015 https://arxiv.org/abs/1506.01497
- YOLO, 2015 https://arxiv.org/abs/1506.02640
- YOLOv2, 2016 https://arxiv.org/abs/1612.08242
- Feature Pyramid Networks, 2017 https://arxiv.org/pdf/1612.03144.pdf
- DETR, 2020 https://arxiv.org/abs/2005.12872